Advanced Database System

Relational Algebra

Agenda:

What is Relational Algebra

Fundamental Operations

* Select
* Project
* Union
* Set different
* Cartesian product
* Rename

Relational Algebra

- It is a **procedural** query language
- RA is a **set of operations** on relation(s)
- It is a set of **algebraic operations**
- Input : One or more relations
- Output : A relation
- It provides a **theoretical foundation** for relational databases.
- **Structured Query Language** (SQL)
- It allows us to understand database operations in more detail and motivate us to write **optimized queries**.

Select Operation (σ)

It selects tuples that satisfy the given predicate from a relation.

- **Notation** – $\sigma_p(r)$

Where **σ** stands for selection predicate and **r** stands for relation. *p* is prepositional logic formula which may use connectors like **and, or,** and **not**. These terms may use relational operators like – =, ≠, ≥, < , >, ≤.

Example

$\sigma_{subject = "database"}$(Books)

Output – Selects tuples from books where subject is 'database'.

$\sigma_{subject = "database" \text{ and } price = "450"}$(Books)

Output – Selects tuples from books where subject is 'database' and 'price' is 450.

$\sigma_{subject = "database" \text{ and } price = "450" \text{ or } year > "2010"}$(Books)

Output – Selects tuples from books where subject is 'database' and 'price' is 450 or those books published after 2010.

Exercises (Select Operation (σ))

1: Write an RA expression to find all the instructors working in “Finance” department

2: Find all instructors with salary greater than 87000

3: Find all instructors who are working in “Finance” department and drawing the salary greater than 87000

INSTRUCTOR

ID	Name	Dept_Name	Salary
10101	John	Biology	65000
12121	Robin	Computer Science	90000
25252	Alya	Electrical	40000
26589	Yusuf	Finance	95000
54789	Ravi	Music	60000
78787	Raj	Physics	87000
87458	Jayant	History	75000
76985	Patrick	Computer Science	89000
12547	Neil	Finance	80000

Project Operation (∏)

It projects column(s) that satisfy a given predicate.

Notation – $\prod_{A1, A2, An} (r)$

Where A_1, A_2 , A_n are attribute names of relation **r**.

Duplicate rows are automatically eliminated, as relation is a set.

Example

$\Pi_{\text{subject, author}}$ (Books)

Selects and projects columns named as subject and author from the relation Books.

Exercises (Project Operation (∏))

1: List all instructors' ID, name, and salary.

2: Find the name of all instructors in the Computer Science department.

3: Find all the names of all instructors with a Salary > greater than 40,000.

INSTRUCTOR

ID	Name	Dept_Name	Salary
10101	John	Biology	65000
12121	Robin	Computer Science	90000
25252	Alya	Electrical	40000
26589	Yusuf	Finance	95000
54789	Ravi	Music	60000
78787	Raj	Physics	87000
87458	Jayant	History	75000
76985	Patrick	Computer Science	89000
12547	Neil	Finance	80000

Union Operation (∪)

It performs binary union between two given relations and is defined as −

$r \cup s = \{ t \mid t \in r \text{ or } t \in s\}$

Notation − r U s

Where **r** and **s** are either database relations or relation result set (temporary relation).

For a union operation to be valid, the following conditions must hold −

- **r**, and **s** must have the same number of attributes.
- Attribute domains must be compatible.
- Duplicate tuples are automatically eliminated.

Example

Π_{author} (Books) $\cup$ Π_{author} (Articles)

Output – Projects the names of the authors who have either written a book or an article or both.

Set Difference (−)

The result of set difference operation is tuples, which are present in one relation but are not in the second relation.

Notation − r − s

Finds all the tuples that are present in **r** but not in **s**.

Example

Π_{author} (Books) − Π_{author} (Articles)

Output – Provides the name of authors who have written books but not articles.

Relational Algebra (Set Difference Operation)

4. Set Difference (–)

Example 1: List all customer names those who have a deposit account but not availed loan.

DEPOSITOR

Cu_Name	Acc_No
Tom	A-101
Amy	A-502
Rose	A-304
John	A-689

BORROWER

Cu_Name	Loan_No
John	L-201
Smith	L-658
Rose	L-254
Jack	L-547

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Relational Algebra (Set Difference Operation)

4. Set Difference (–)

Example 2: Find all the courses taught in the Fall 2009 semester but not in Spring 2010.

Relational Algebra (Set Difference Operation)

Example – University Database

Instructor (ID, Name, Dept_Name, Salary)

Course (Course_ID, Title, Dept_Name, Credits)

Department (Dept_Name, Building, Budget)

Section (Course_ID, Sec_ID, Semester, Year, Building, Room_No, Time_slot_ID)

Teaches (ID, Course_ID, Sec_ID, Semester, Year)

Student (ID, Name, Dept_Name, Tot_Cred)

Advisor (S_ID, I_ID)

Takes (ID, Course_ID, Sec_ID, Semester, Year, Grade)

Classroom (Building, Room_Number, Capacity)

Time_Slot (Time_Slot_ID, Day, Start_Time, End_Time)

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Relational Algebra (Set Difference Operation)

4. Set Difference (–)

Example 2: Find all the courses taught in the Fall 2009 semester but not in Spring 2010.

Solution:

To find the set of all courses taught in the Fall 2009 semester, we write:

$$\Pi_{Course_id} (\sigma_{Semester = \text{"Fall"} \wedge year = 2009} (SECTION))$$

To find the set of all courses taught in the Spring 2010 semester, we write:

$$\Pi_{Course_id} (\sigma_{Semester = \text{"Spring"} \wedge year = 2010} (SECTION))$$

To answer the query, we need the minus of these two sets:

$$\Pi_{Course_id} (\sigma_{Semester = \text{"Fall"} \wedge year = 2009} (SECTION)) - \Pi_{Course_id} (\sigma_{Semester = \text{"Spring"} \wedge year = 2010} (SECTION))$$

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Cartesian Product (X)

Combines information of two different relations into one.

Notation – r X s

Where **r** and **s** are relations and their output will be defined as –

$r \times s = \{ q\ t \mid q \in r \text{ and } t \in s \}$

Example

$\sigma_{\text{author = 'tutorialspoint'}}$(Books X Articles)

Output – Yields a relation, which shows all the books and articles written by tutorialspoint.

5. Cartesian Product (X)

Example: Perform Depositor X Borrower.

DEPOSITOR

Cu_Name	Acc_No
Tom	A-101
Rose	A-304

BORROWER

Cu_Name	Loan_No
John	L-201
Smith	L-658
Rose	L-254
Jack	L-547

DEPOSITOR X BORROWER

Cu_Name	Acc_No	Cu_Name	Loan_No
Tom	A-101	John	L-201
Tom	A-101	Smith	L-658
Tom	A-101	Rose	L-254
Tom	A-101	Jack	L-547
Rose	A-304	John	L-201
Rose	A-304	Smith	L-658
Rose	A-304	Rose	L-254
Rose	A-304	Jack	L-547

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Relational Algebra (Cartesian Product Operation)

5. Cartesian Product (X)

Example : Find the names of all instructors in the Physics department together with the course id of all courses they taught.

Solution:

$\sigma_{Dept_Name = \text{"Physics"}}$ (INSTRUCTOR X TEACHES)

$\sigma_{Instructor.ID = Teaches.ID}$ ($\sigma_{Dept_Name = \text{"Physics"}}$ (INSTRUCTOR X TEACHES))

$\Pi_{name,\ course_id}$ ($\sigma_{Instructor.ID = Teaches.ID}$ ($\sigma_{Dept_Name = \text{"Physics"}}$ (INSTRUCTOR X TEACHES)))

[or]

$\Pi_{name,\ course_id}$ ($\sigma_{Instructor.ID = Teaches.ID}$ (($\sigma_{Dept_Name = \text{"Physics"}}$ (INSTRUCTOR) X TEACHES))

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Rename Operation (ρ)

The results of relational algebra are also relations but without any name. The rename operation allows us to rename the output relation. 'rename' operation is denoted with small Greek letter **rho** ρ.

Notation − $\rho_x(E)$

Where the result of expression **E** is saved with name of **x**.

Example

$\sigma_{\text{author = 'tutorialspoint'}}$(Books X Articles)

Output – Yields a relation, which shows all the books and articles written by tutorialspoint.

References

- https://www.tutorialspoint.com/dbms/relational_algebra.htm
- Neso Academy Youtube Channel
- Elmasri, R. and S. Navathe. Fundamentals of Database. Seventh Edition.